



EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2025	B.TECH	IIT Kharagpur	8.69 / 10
2021	AISSE (XII) CBSE	MP and EV school	93.5%
2019	AISSE (X) CBSE	Visakha Valley School	91%

INTERNSHIPS

- MITACS GRI | University of Alberta | Canada** **May 2024 - Aug 2024**
- Developed a conceptual design of a **new age Nanosatellite** integrated into a drone sized launch vehicle functioning in very low earth orbit
 - Analysed the concept feasibility for **3+** propulsion engine integrated design by simulating trajectory in **Python** and **MATLAB** environment
 - Solicited for the **Diana-NATO grant** to demonstrate the feasibility of the innovative concept of spaceplanes for the first time ever globally
- Research Intern | Space Applications Center(SAC) | ISRO** **May 2023 - July 2023**
- Reduced the power consumption of nanosatellites upto **30%** by developing an efficient attitude control system manoeuvring orientation
 - Achieved a minimum stabilization time of **60** seconds using **Python/MATLAB** simulations for high performing detumbling of nanosatellites
 - Designed and implemented power management and distribution system (**EPS**) in various syusbsystems for efficient power utilization
- Data Analyst Intern | Pinvest Inc.** **June 2024 - Aug 2024**
- Analyzed property data for **8** major Indian cities extracting housing price data to create hedonic model for accurate price prediction
 - Extracted Residential housing data by scraping **10+ yearly real estate** reports using libraries like pytesseract for useful data extraction
 - Modeled the housing price index adjusted to market price for **5+** years for price prediction and trading it as an **ETF**(Exchange Traded Fund)
- EU Analytics Intern | WNS Global** **May 2023 - June 2023**
- Acquired **2+** new clients by developing an ESG risk score methodology with **20+** numerical metrics in **3 ESG pillars** using MSCI Ratings
 - Derived a numerical based scoring moethod using **4+ sentimental factors** developing a new method for assessing governance metrics
 - Devised **3 new formulae** under environmental and social pillar to assess time series risk score utilizing statistical based scaling method
- Finance Intern | MBA HUB** **Dec 2022 - Jan 2023**
- Formulated a stock report of **3** companies including Reliance and Axis Bank based on fundamental analysis and **candlestick chart data**
 - Generated long or short **trade signals** by analyzing support &resistance lines, identifying trade volumes and utilizing moving averages
 - Developed long term investment plans by conducting market research and fundamental analysis of **3+** types of financial reports

COMPETITION/CONFERENCE

- Cricket Winner Prediction Model | American Express Decision Science Challenge** **May 2024 - Jun 2024**
- Ranked in the top **20** out of the **450+** submissions by creating **50+** new features using feature engineering for the prediction model
 - Achieved a peak accuracy of **60%** using **4 boosting algorithms** GBM, LightGBM, CatBoost, XGBoost, and their ensemble on the datasets
 - Streamlined **feature validation tests** (Chi-squared test), fine-tuning techniques and **L1/L2 regularization**, to optimize accuracy scores

PROJECTS

- Collision of Smoke Vortex Rings | Guided by Prof.Sandeep Saha Sir** **Feb 2022 - Apr 2023**
- Engineered an airtight acrylic chamber to generate circular smoke vortex rings, extending research from water to the air domain
 - Designed and 3D printed funnels for variable-sized rings and synchronized their impacts using **Arduino** and **servo** motors for control
 - Investigated the interaction **time variations** for head-on and side-on collisions across diverse combinations of radii and applied impulses
 - Submitted an abstract for consideration to the **EFDC1 Conference, Aachen** and a video to the **Travelling Gallery of Fluid Motion, Utah**

AWARDS AND ACHIEVEMENTS

- Got Featured in **The Indian Express'** series 'Life in an IIT' and was showcased/featured on IIT Kharagpur's official social media handles
- Secured Department rank of **11** out of the **77(B.Tech + Dual Degree)** students in the department of aerospace engineering, IIT Kharagpur
- Scored **95.09** percent in the PDR round of the prestigious Cansat USA'23 competition organized by American Astronomical Society(AAS)
- Was selected for **2024 HPAIR(Harvard Project for Asian and International Relations)** Conference Virtual Summit, among thousands
- Secured All India Rank of **4448** in highly competative exam named JEE Advanced 2021(out of 1,42,000 candidates)

SKILLS AND EXPERTISE

Languages: C++, Python(Libraries: Pandas, Numpy, Scikit-Learn, NLTK,Tensorflow), Matplotlib, HTML, MATLAB, Prompt Engineering
Software: MATLAB, Jupyter notebook, Solidworks, Ansys(Fluent), MS Office, Adobe Photoshop, Adobe illustrator, Simulink

COURSEWORK INFORMATION

Programming courses: Programming and Data Structures|Artificial Intelligence Foundations and its Applications|Systems lab(Simulink)
Mathematical Courses: Linear Algebra,Numerical and Complex analysis|Transform Calculus|Numerical solution of ODE and PDE
Core Courses: Flight Vehicle controls|Fundamental of embedded control and software|High Speed Aerodynamics|Mechanics of Flight

EXTRA CURRICULAR ACTIVITIES

- Earned a KUKKIWON Certified second **DAN Black Belt** in Taekwondo which is a South Korean martial art in South Korea - 2017.
- Secured a **Silver Medal** in Taekwondo representing the state of Gujarat at the Students Olympic National Games in 2014 and 2015.
- Claimed a **Bronze Medal** in Taekwondo at the 62nd National School Games held in Delhi in 2016/2017, representing the state of Gujarat.
- Competed at the National Federation Cup in Nashik in 2017, achieving a **Bronze Medal** in Taekwondo representing the state of Gujarat.
- Participated in GC **weight lifting** representing Pandit Madan Mohan Malaviya Hall in under 77 kg category securing **5th position** among all
- Selected as an **SWG Mentor** for IIT Kharagpur's freshman batch of 2024, guiding them through campus challenges and opportunities